MILITARY SPECIFICATION SHEET

ELECTRON TUBE, KLYSTRON

1/ TYPE DOD-002

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Reflex, CW, tunable frequency range 4,100 to 4,450 MHz, nominal power output 4 mW.

ABSOLUTE RATINGS:

Parameter:	Ef	Ers	Er	Ik	Ehk
Unit:	V	Vdc	Vdc	mAdc	Vdc
Maximum:	6.6	330	-400	35	50
Minimum:	6.0	---	---	---	---

PHYSICAL CHARACTERISTICS:

Dimensions: See figure 1
Cathode: Unipotential, oxide coated

TEST CONDITIONS: See note 1

Parameter:	Ef	Ers	Er	F
Unit:	V	Vdc	Vdc	MHz
Test condition 1:	6.3	300	-40 to -155	F3
Test condition 2:	6.0	250	-42 to -82	F1
Test condition 3:	6.0	250	-67 to -87	F2
Test condition 4:	6.0	250	-69 to -109 (See note 2)	F3

FREQUENCY	
F	MHz
F1	4,100
F2	4,300
F3	4,450

GENERAL:

Qualification - Required
Holding period - 72 hours
1/ Formerly type QK-381

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METHOD	REQUIREMENT OR TEST	NOTES	TEST	CONDITIONS	SYMBOL	LIMITS		UNIT
						MINIMUM	MAXIMUM	
	Qualification inspection							
1031	Nonoperation vibration	7	-	F = 50 Hz, 10G; t = 120	---	---	---	---
4027	Temperature coefficient	7	-		ΔF ΔT	-0.10	-0.05	MHz C
	Quality conformance inspection, part 1	3	-					
4214	Cathode emission	-	1	Ef = 5.8 V	ΔIk Ik	---	15	
1256	Electrode current (cathode)	1	1		Ik	---	32	mAdc
4250	Power output	5	2, 3, 4	VSWR = 1.1:1	Po	3	---	mW
4213	Reflector voltage	- -	2 4		Er Er	-42 -69	-82 -109	Vdc Vdc
4229	Total reflector current	6	4		Ir	---	7.0	μAdc
4229	Reflector leakage current	6	4		Ir	---	5.0	μAdc
4229	Reflector gas current	6	4		Ir	---	2.0	μAdc
---	Tuner setting	4	3		---	---	---	---
	Quality conformance inspection, part 2							
4289	Heater current, nonoperating	-	-	Ef = 6.3 V	If	410	470	mA
1211	Insulation of electrodes	-	-	Ers = 300 Vdc; tube cold	Rk-rs Rh-rs	2.0 2.0	--- ---	Meg Meg
4270	Heater-cathode leakage	-	-	Ehk = +45 Vdc	Ihk	0	100	μAdc
4280	Electronic tuning range	5	1	Er 50% max Po	ΔF	30	---	MHz
4231	Electronic tuning hysteresis	-	1		Ratio	---	0.05	---
	Quality conformance inspection, part 3							
---	Life test	-	1	Group B	t	500	---	hrs
---	Life-test end point:							
4250	Power output	5	2, 3, 4		Po	2.5	---	mW

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OTES:

1. The TUT shall be fixed firmlv in a socket bv clamps which make contact at the base and adjacent platform onlv. The measurements on the tubes in an oscillating state shall be made with the output line coupled through a connector into a 50-ohm lossv line operating into a load adjusted to 50 ohms at 4,300 MHz.

2. Reflector voltage shall be adjusted to the value within the specified limits which is necessarv to obtain maximum power output.

3. Unless otherwise specified, the AQL for all tests listed under qualitv conformance inspection, part 1, shall be 1.0, inspection level II.

4. When the TUT is set at F2 ± 0.5 percent, the tuner bow shall be opened 2-1 4 to 3-3 4 turns from a closed position.

5. The TUT shall meet the requirements of this test over the operating band, however, the test need be conducted onlv at the designated frequency or frequencies.

6. This test to be performed at the conclusion of the holding period.

7. Revision letters are not used in this revision to identifv changes with respect to the previous issue, due to the extensiveness of the changes.

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Review activities:
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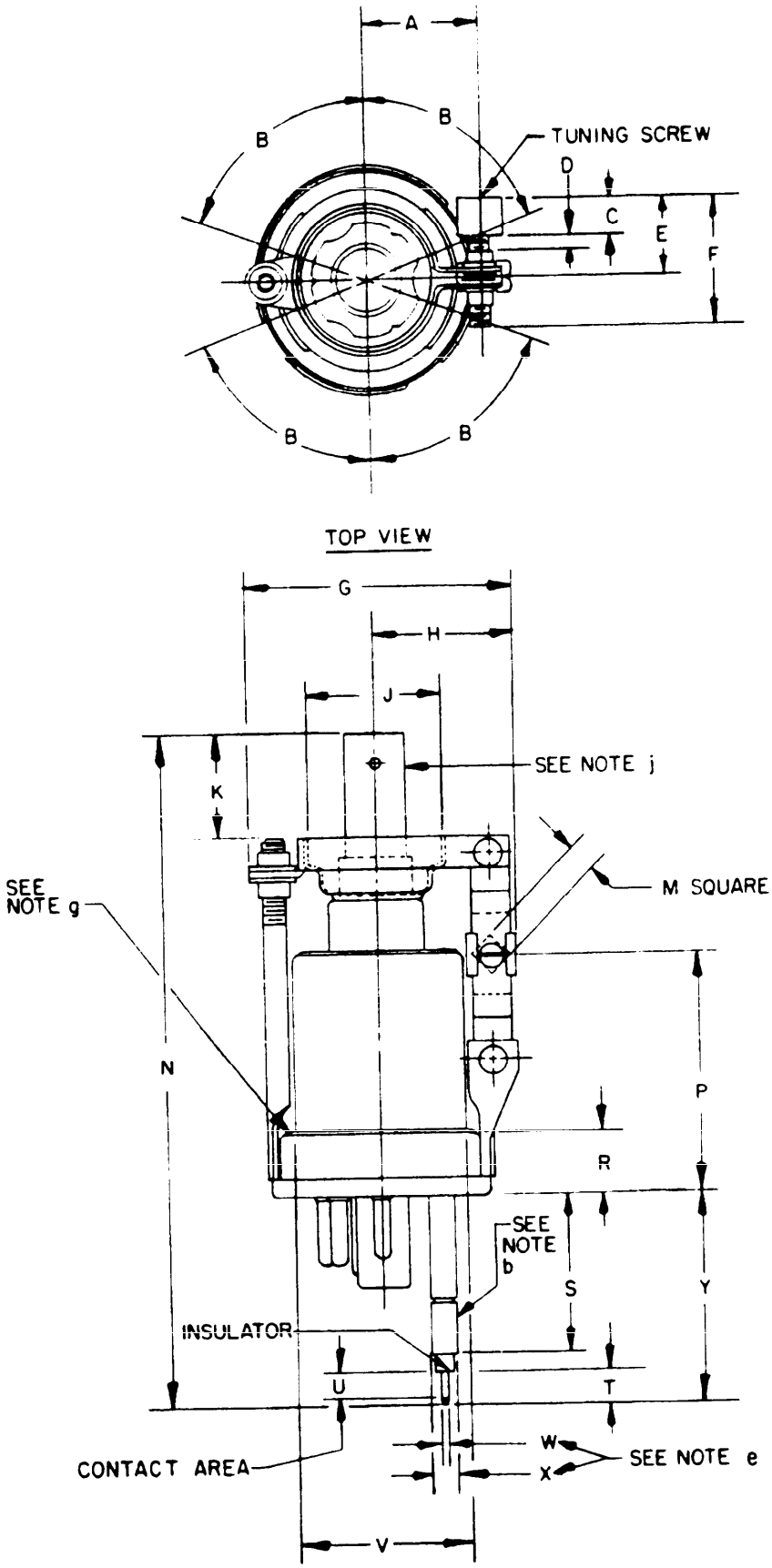
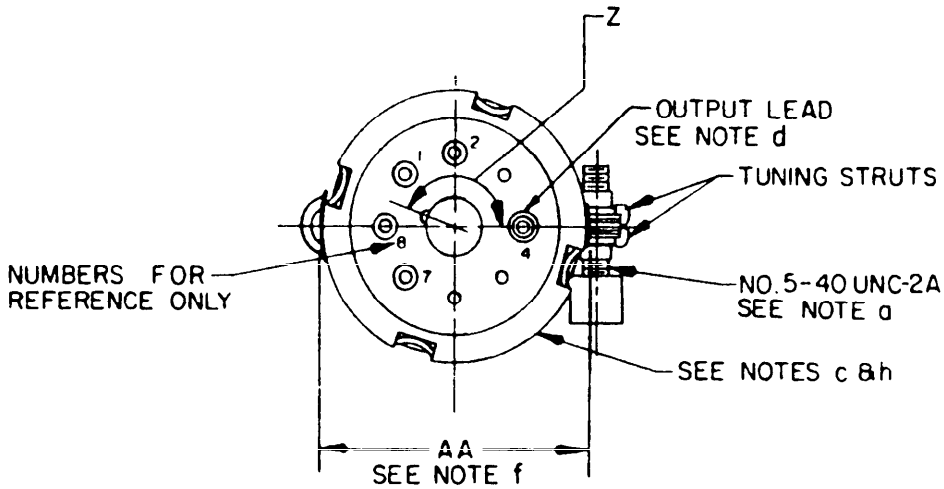


FIGURE 1. Outline drawing of tube type DOD-002

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BOTTOM VIEW

PIN CONNECTIONS	
PIN NO.	ELEMENT
1	sh
2	h
4	OUTPUT LEAD
7	h
8	GUN
CAP	r

FIGURE 1. Outline drawing of tube type DOD-002 - Continued.

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses			
	Minimum		Maximum	
Quality conformance inspection, part 1 (see note k)				
A	.657	(16.69)	.719	(18.26)
B			[illegible]	
C	.203	(5.16)	.234	(5.94)
D			.135	3.43
E	.503	(12.78)	.535	13.59
F	.804	(20.42)	.835	(21.21
J	.767	(19.48)	.797	(20.24)
K	.531	(13.49)	.593	(15.06
M	.182	(4.62)	.192	(4.88)
P	1.325	(33.66)	1.425	(36.20)
R	.330	(8.38)	.400	(10.16
S	.875	(22.23)	.984	(24.99)
T	.172	(4.37)	.203	5.16)
V	1.000 DIA	(25.40)	1.015 DIA	(25.76)
W	.035 DIA	(.89)	.045 DIA	(1.14)
X	.135 DIA	(3.43)	.145 DIA	(3.68)
Y	1.203	(30.56)	1.234	(31.34)
Z	150° 30'		160° 30'	
AA	1.271	(32.28)	1.312	(33.32)
Quality conformance inspection, part 2				
G			1.609	40.87)
H			.359	21.32)
N			3.950	100.33)
[illegible]	.156	3.96)		

NOTES:

a. The tuning screw shall be lubricated with oildag or equal non-corrosive lubricant. It shall be capable of being operated smoothly through its entire range without perceptible binding.
b. Nickel (30 MSI silver plate permissible).
c. The base shall be capable of being inserted in a gage 1.219 (30.96 mm) having 4 holes .250 (6.35 mm) deep from the top of gage whose diameters are .103 (2.62 mm) for the contact pins. Remaining portion of hole to be clearance approximately .016 (.41 mm) larger in diameter and a fifth hole whose diameter is .160 (4.06 mm) by 1.219 (30.96 mm) deep for the output lead. All holes located on the true centers. Also, a center hole having the contour of the pilot but with a clearance of .002 (.05 mm) over the maximum diameter.
d. Applies to centerline passing through center of output lead and midway between tuning struts.
e. Shall be concentric with each other within .010 (.25 mm).
f. This dimension applies to wafer portion of base only.
g. Area free for mounting.
h. Standard octal base insert modified for output lead only.
j. Couple with BNC connector jack UG-89D/U conforming to drawing RED 49132.
k. Unless otherwise specified, the AQL for all tests listed under quality conformance inspection, part 1, shall be 1.0 percent, inspection level of II.

FIGURE 1. Outline drawing of tube type DOD-002. - Continued.

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